

# WASP-36b

R-2438

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-36\_130118 · created 2026-08-22T01:34:53+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456310.84735 +/- 0.00047 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.167 +/- 0.019                   |
| Transit depth (Rp/Rs)^2                           | 2.78 +/- 0.64 [%]                 |
| Orbital Inclination (inc)                         | 83.59 +/- 1.35                    |
| Airmass coefficient 1 (a1)                        | 1.246 +/- 0.018                   |
| Airmass coefficient 2 (a2)                        | -0.157 +/- 0.051                  |
| Scatter in the residuals of the lightcurve fit is | 1.46 %                            |
| Best Comparison Star                              | #2 - [116, 304]                   |
| Optimal Aperture                                  | 3.23                              |
| Optimal Annulus                                   | 7.08                              |
| Transit Duration (day)                            | 0.0806 +/- 0.0085                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.167 ± 0.019 vs 0.1368 (deviation 1.32σ) |
| Duration fit vs published | 0.0806 d vs 0.0758 d (deviation 0.46σ)    |
| Mid-time O–C              | -1.1 min                                  |
| Depth SNR                 | 7.3                                       |
| Residual scatter          | 1.46 %                                    |

## Light curve

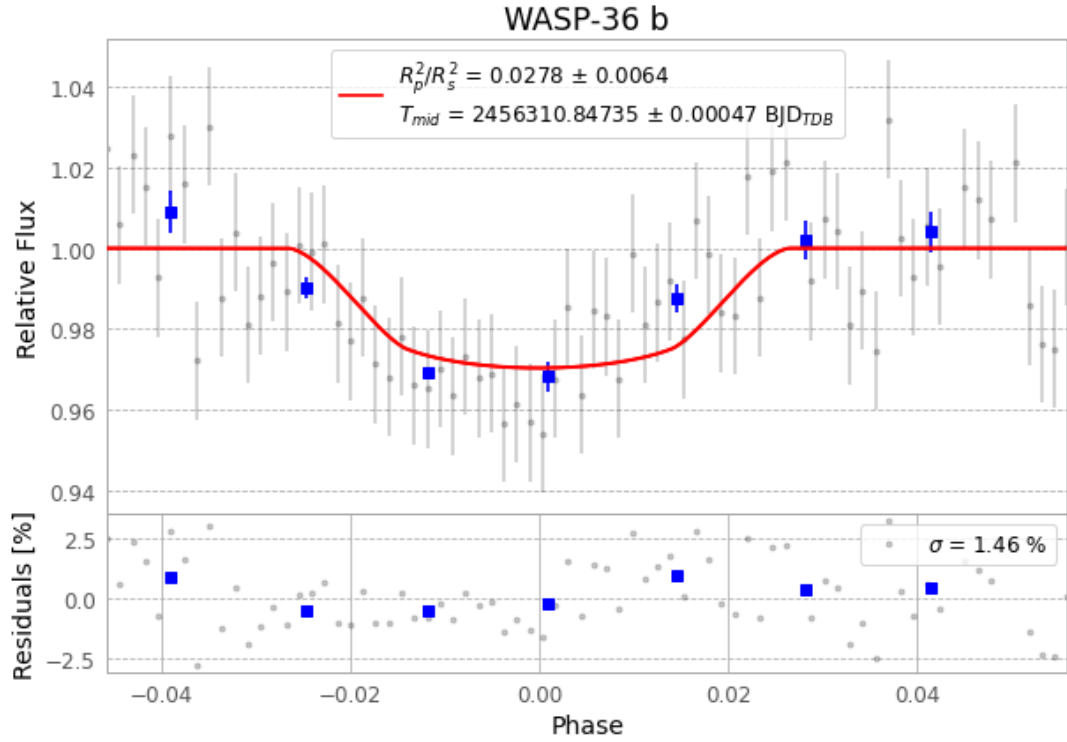


Figure 1 — FinalLightCurve\_WASP-36 b\_2013-01-17.png (SHA-256 578140164e995de7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [257, 286], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 c8d1514f72f7005e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

76 FITS frames (50 MB), WASP-36130118063012.FITS → WASP-36130118101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-130118.log (048dc3fa6029...), FinalLightCurve\_WASP-36 b\_2013-01-17.png (578140164e99...), FinalLightCurve\_WASP-36 b\_2013-01-17.csv (90bcf761afa1...)

## Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 01:34Z.